MILITARY SPECIFICATION

CONVERTER, INTERFACE UNIT, CV-3507/USQ-74

This specification is approved for use by the Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the design, construction and performance requirements of the converter interface unit (CIU), CV-3507/USQ-74, (hereinafter referred to as the CIU), a unit of the LINK 11 AN/USQ-74 data terminal set (DTS). The CIU shall receive two analog signals from a LINK 11 radio set, derive two signals, one in-phase and one quadrature from each analog signal, and convert each of these derived signals to digital form. The resulting digital signals are transferred by the CIU to an AN/UYK-20 computer. The CIU shall accept digital signals from the AN/UYK-20 computer and convert these digital signals to analog signals for modulating the radio frequency signals transmitted by a LINK 11 radio set. The CIU shall transfer the analog signals to the radio set and send keyline control signals to the radio set.

2. REFERENCED DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-C-5015	Connectors, Electrical, Circular Threaded, AN Type, General Specifications For
MIL-M-7793	Meter, Time Totalizing
MIL-P-15024	Plates, Tags And Bands For Identification Of Equipment
MIL-E-16400	Electronic, Interior Communication And Navigation Equipment, Naval Ship And Shore: General Specification For
MIL-E-17555	Electronic And Electrical Equipment, Accessories, And Repair Parts, Packaging And Packing Of
MIL-E-19207	Fuseholder, Extractor Post Type, Blown Fuse Indicating And Nonindicating, General Specification For
MIL-C-39012	Connectors, Coaxial, Radio Frequency, General Specification For
MIL-H-46855	Human Engineering Requirements For Military Systems, Equipment And Facilities

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Electronic Systems Command (ELEX-81111), Washington, DC 20360, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

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STANDARDS

MILITARY

MIL-STD-12	Abbreviations For Use On Drawings, Specifications, Standards, And In Technical Documents
MIL-STD-108	Definitions Of And Basic Requirements For Enclosures For Electric And Electronic Equipment
MIL-STD-109	Quality Assurance Terms And Definitions
MIL-STD-130	Identification Marking Of U.S. Military Property
MIL-STD-189	Rack, Electrical Equipment, 19-Inch And Associated Panels
MIL-STD-415	Test Provisions For Electronic Systems And Associated Equipment, Design Criteria For
MIL-STD-454	Standard General Requirements For Electronic Equipment
MIL-STD-461	Electromagnetic Emission And Susceptibility Requirements For The Control Of Electromagnetic Interference
MIL-STD-462	Electromagnetic Interference Characteristics, Measurement Of
MIL-STD-471	Maintainability Demonstration
MIL-STD-701	Lists Of Standard Semiconductor Devices
MIL-STD-781	Reliability Design Qualification And Production Acceptance Tests: Exponential Distribution
MIL-STD-810	Environmental Test Methods
MIL-STD-965	Parts Control Program
MIL-STD-1364	Standard General Purpose Electronic Test Equipment
MIL-STD-1397	Input/Output Interfaces, Standard Digital Data, Navy Systems
DoD-STD-1399, Section 300	Interface Standard For Shipboard Systems, Electric Power, Alternating Current, Section 300
MIL-STD-1472	Human Engineering Design Criteria For Military Systems, Equipment And Facilities

HANDBOOKS

MILITARY

MIL-HDBK-241	Design Guide For EMI Reduction In Power Supplies

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

PUBLICATIONS

FEDERAL

Code Of Federal Regulations Title 29, Part 1910 Subpart 2	Occupational Safety And Health Standards

(Applications for copies should be addressed to Department of Labor, Occupational Safety and Health Agency (OSHA), Washington, DC 20210.)

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NAVAL ELECTRONIC SYSTEMS COMMAND (NAVELEX)

0967-LP-598-1010 Series	Data Processing Set, AN/UYK-20(V) Technical Manual, Operation and Maintenance With Parts List (Volumes 1-6)

NAVAL PERSONNEL (NAVPERS)

18068	Manual of Navy Enlisted Manpower and Personnel Classifications and Occupational Standards

NAVAL OCEAN SYSTEMS CENTER (NOSC)

TM 344, Change 1 of June 1979	Type A System Specification for LINK 11 AN/USQ-74 Data Terminal Set (DTS)
TM 348, Change 1 of June 1979	Program Performance Specification for LINK 11 AN/USQ-74 Data Terminal Set (DTS)

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General requirements. The CIU, a part of the LINK 11 AN/USQ-74 DTS, shall be in accordance with MIL-E-16400, except as hereinafter specified. The CIU shall provide the interface between any LINK 11 radio set specified in 3.1.2.1 and the AN/UYK-20 computer (NAVELEX publication 0967-LP-598-1010 series). The CIU shall perform quadrature phase shifting, sampling, A/D conversion, D/A conversion, filtering, signal amplification, timing and control as specified herein to conform to the requirements of this interface. Receive and transmit, digital and analog, channels shall be separate. The functional block diagram and signal flow in the CIU shall be in accordance with FIGURE 1. FIGURE 2 is a system block diagram showing the LINK 11 application.

3.1.1 First article. When specified, a sample shall be subjected to first article inspection (see 4.3 and 6.2).

3.1.2 System interoperability and compatibility. The CIU shall be interoperable and compatible with the system and individual performance characteristics and signal waveforms described in this specification.

3.1.2.1 Compatibility with radio sets. The CIU shall be compatible with the following shipboard radio sets: AN/SRC-16, AN/SRC-23, AN/SRC-31, AN/URC-75, AN/URC-85, AN/URT-23C and R-1051F/URR or R-1051G/URR combination, AN/URC-81 and T-1322/SRC, R-1903/UR and the AN/URC-97 radio set, AN/WSC-3(V)6/7 and AN/URC-93(V)1 combination. In this document compatibility shall be interpreted to mean that (a) the CIU shall accept any LINK 11 audio output signals from these radio sets while operating as specified herein, and (b) the CIU shall provide audio input signals sufficient to drive any one of these radio sets while operating as specified herein.

3.1.3 Operation with Government furnished equipment. The CIU shall operate with the Government furnished equipment (GFE), AN/UYK-20 computer, AN/USQ-69 terminal, the Government-furnished software for the AN/UYK-20 and the AN/USH-26 cartridge magnetic tape unit in accordance with Naval Ocean Systems Center (NOSC) TM 344 and TM 348, both with Change 1 of June 1979.

3.1.4 Modifications. The design of the CIU shall require no modifications to the AN/UYK-20 computer or any of the radio sets of 3.1.2.1 with which the CIU must be compatible.

3.1.5 Interfaces. The CIU electrical power and system analog and digital signal interfaces shall be electrically and physically compatible with corresponding interfaces in the GFE.

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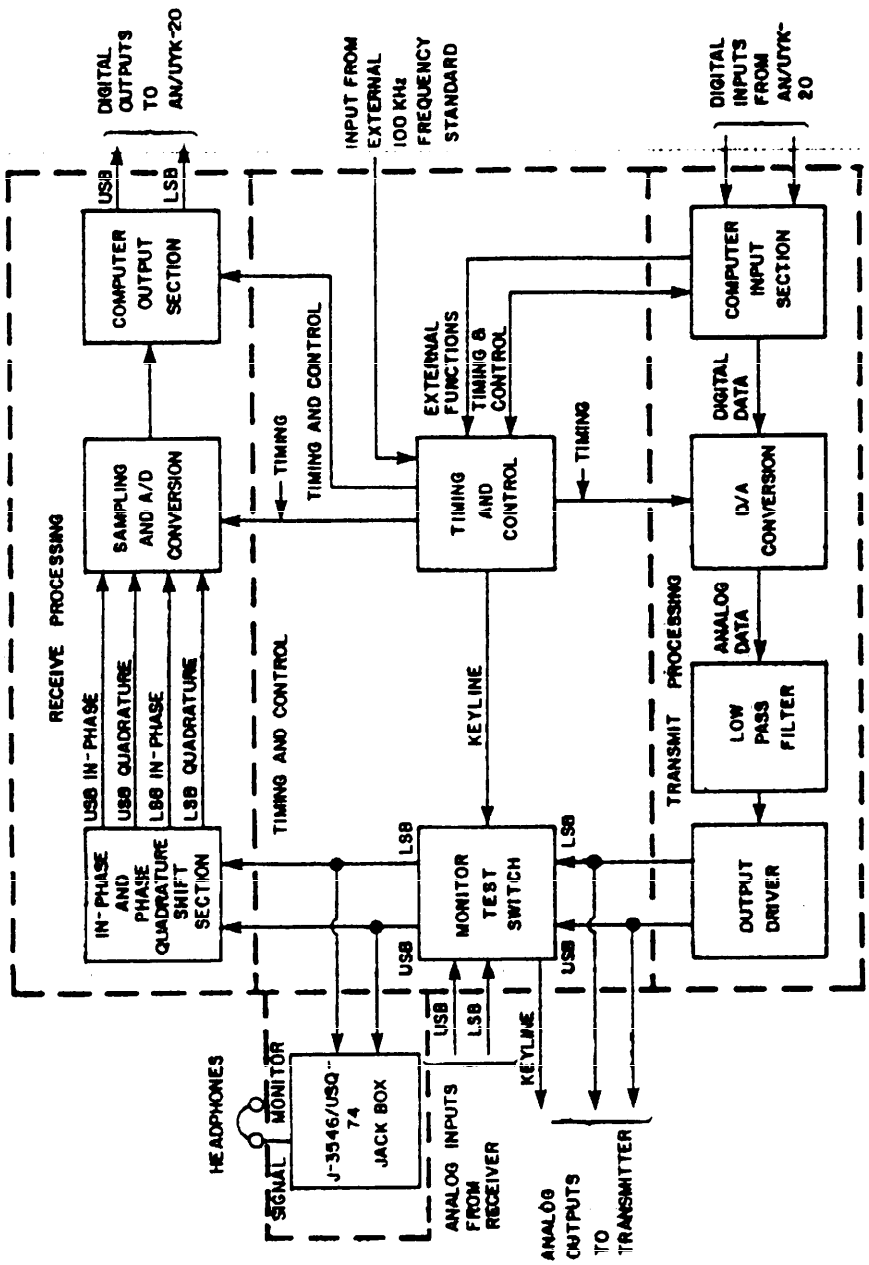


FIGURE 1. CV-3507/USQ-74 converter interface unit functional block diagram.

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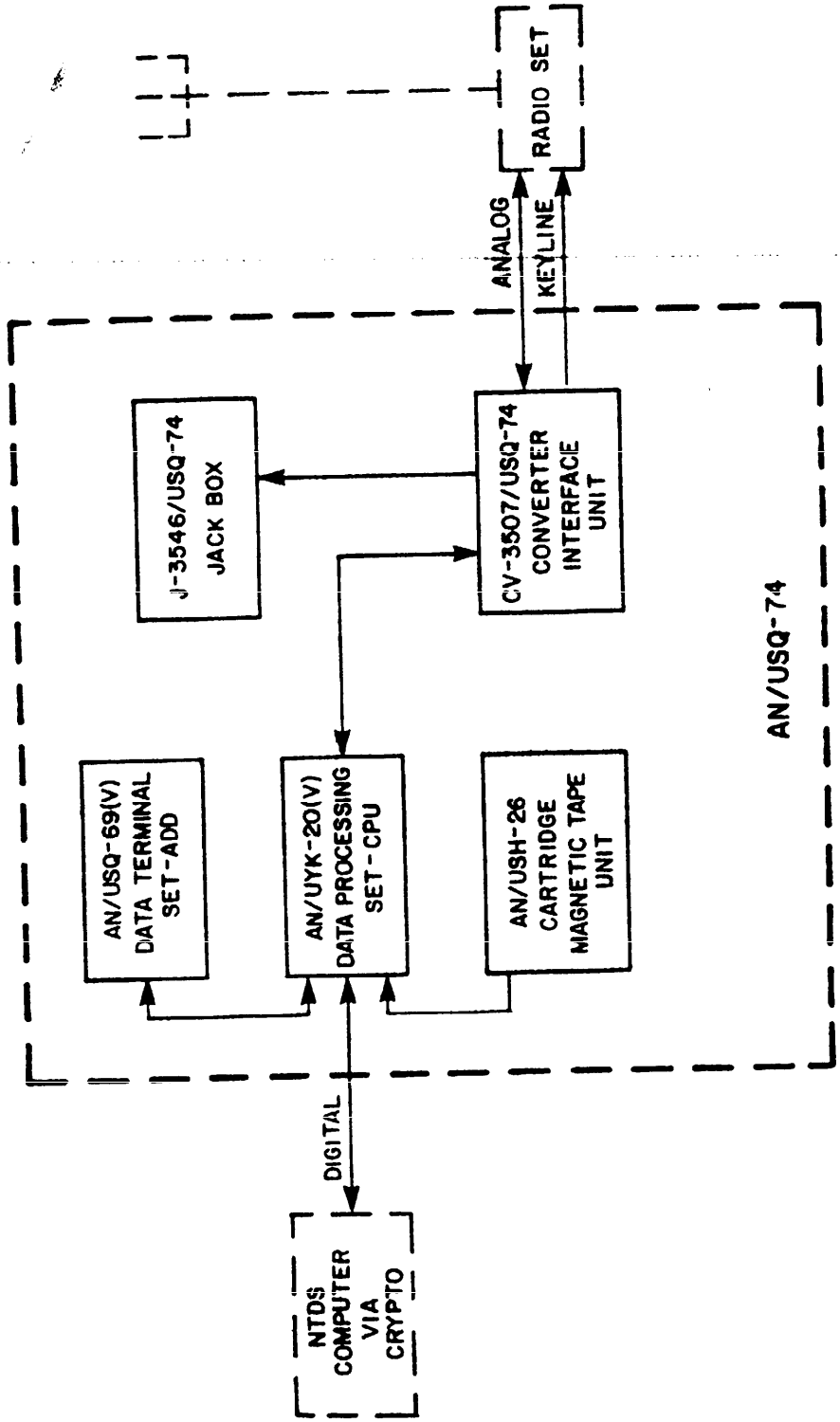


FIGURE 2. AN/USQ-74 system diagram showing CV-3507/USQ-74 in typical LINK 11 application.

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3.2 Characteristics.

3.2.1 Performance characteristics.

3.2.1.1 Receive processing. The CIU shall receive analog signals from the radio set, process these signals as specified herein, and transfer the processed digital signals to the AN/UYK-20 computer for further processing.

3.2.1.1.1 Input. The CIU shall accept and simultaneously process two audio frequency signals, on separate channels, with each signal of up to 12 volts peak-to-peak maximum from a 600-ohm source. These two signals shall be the upper sideband (USB) and the lower sideband (LSB) signals received and translated to baseband by the radio set. The input transformer impedance for each of the two radio input receive channels shall be 600 ohms ±10 percent, resistive and balanced over the frequency range 500 to 3050 Hz. The electrical balance shall be sufficient to suppress the longitudinal currents at least 40 decibels (dB) below a 0 dBm (decibel referenced to one milliwatt) level input. The CIU shall not be damaged by input signals of up to 20 volts peak-to-peak.

3.2.1.1.2 Quadrature phase shifting. Each of the two input signals shall be quadrature phase shifted by the CIU. The in-phase and quadrature phase shift section shall have four outputs: an in-phase USB output, a quadrature phase USB output, an in-phase LSB output, and a quadrature phase LSB output. The phase characteristics of the in-phase and quadrature phase shift section shall be as follows:

a. The difference between the phase response of the in-phase USB channel and the phase response of the in-phase LSB channel shall be less than 10 degrees for each tone frequency in the 500-Hz to 3050-Hz band.
b. The phase response of the quadrature phase USB channel shall be 90 degrees, ±4 degrees, relative to the phase response of the in-phase USB channel for each frequency in the 500-Hz to 3050-Hz band.
c. The phase response of the quadrature phase LSB channel shall be 90 degrees, ±4 degrees, relative to the phase response of the in-phase LSB channel for each frequency in the 500-Hz to 3050-Hz band.

The amplitude response of all four in-phase and quadrature phase shift section channels shall be equal within ±1 dB for each frequency in the 500-Hz to 3050-Hz band. For any two-tone combination within the 500-Hz to 3050-Hz frequency band, the differential time delay between the two tones shall be less than 500 microseconds (µs) for each of the four channels of the in-phase and quadrature shift section.

3.2.1.1.3 Sampling and A/D conversions. Each of the four signals from the in-phase and quadrature phase shift section shall be sampled and A/D converted at a rate of 3520 samples per second with equal sample periods. The composite (four channel) sampling and A/D conversion rate shall be 14,080 samples per second.

3.2.1.1.3.1 Aperture time. The aperture time of the sampling process shall be less than 100 nanoseconds (ns).

3.2.1.1.3.2 Dynamic range. The A/D conversion process shall have a dynamic range of 12 bits (11 bits plus sign bit).

3.2.1.1.3.3 A/D conversion linearity. The A/D conversion process shall be linear to within plus or minus one-half of the least significant bit at 25° Celsius (C) with a temperature coefficient not greater than 35 parts per million of full scale per degree C.

3.2.1.1.3.4 A/D conversion accuracy. The A/D conversion process shall have an accuracy within ±0.3 percent of full scale over the temperature range of 0° C to 50° C.

3.2.1.1.3.5 Sampling timing. The sampling commands for the four in-phase and quadrature phase shift section outputs shall occur within a 10-ns period.

3.2.1.1.3.6 Output. The output of the A/D conversion process to the computer input section shall be in two's complement form.

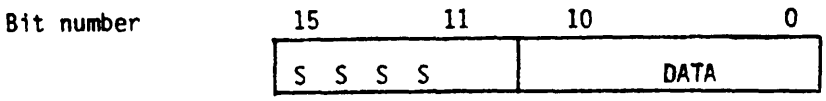
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3.2.1.1.4 Input to the AN/UYK-20 computer from the CIU. The input to the AN/UYK-20 computer from the CIU shall be transferred via two parallel channels. One channel shall be associated with the LSB signal and the other channel shall be associated with the USB signal. The input to the AN/UYK-20 computer from the CIU consists of a series of complex valued samples for each sideband signal. Each complex sample shall consist of one 16-bit word containing the real 12-bit (in-phase) component of the sample and one 16-bit word containing the imaginary 12-bit (quadrature phase) component of the sample. The word containing the real component of each sample shall be transferred to the computer 142 µs before the word containing the imaginary component. A two-word input to the AN/UYK-20 computer from the CIU on a signal channel shall represent one complex sample for one sideband input signal to the CIU.

3.2.1.1.4.1 Computer interface. Each input channel from the CIU shall interface with a standard (I/O) channel (see 3.2.1.2) in the AN/UYK-20 computer. These computer I/O channels ~~provide separate lines for input and output data.~~ These I/O channels shall also be used to transfer output data from the AN/UYK-20 computer for transmit processing (see 3.2.1.2). These I/O channels are described in MIL-STD-1397 and NAVELEX 0967-LP-598-1010 series.

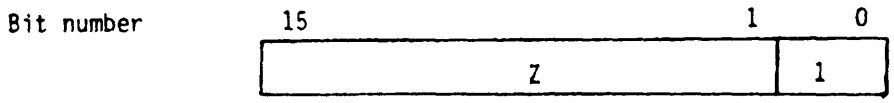
3.2.1.1.4.2 Input data transfer format. The transfer of input data from the CIU to the AN/UYK-20 computer shall be in accordance with MIL-STD-1397, and shall be Type A Navy tactical data system (NTDS) slow, Category I (computer to peripheral), except that the output drivers may switch at a rate in excess of 5.0 volts direct current (VDC) per µs, the Input Data Request (IDR) and Input Data (ID) may be set simultaneously, the External Interrupt Request (EIR) and ID may be set simultaneously and the IDR may be reset by the Input Data Acknowledge (IDA) without delaying until the IDA is fully set.

3.2.1.1.4.3 Input word. The input word to the AN/UYK-20 computer shall be 16 bits which is the same as the word length of the AN/UYK-20. The input word format shall be:



SSSS = Sign filled

3.2.1.1.4.4 Input timing. The timing of the input from the CIU shall be controlled by the CIU which shall provide two complex samples (one on each channel) every 284 µs (see 3.2.1.1.3). If the AN/UYK-20 computer does not sample any input word from the CIU within 71 µs after the word is available, the CIU shall issue a one-time no data interrupt to the AN/UYK-20 on the same channel within 10 µs to indicate lost data. The format for the interrupt word shall be:



Z - Zero filled

After the CIU has issued the no data external interrupt to the AN/UYK-20, there shall be no further input transfers on that channel to the AN/UYK-20 computer until:

a. The channel is cleared by an external function.
b. The channel resets its receive bit in a second external function.

If both channels had been receiving prior to this interrupt, both must follow steps a. and b. and the procedures for USB and LSB channel operation in accordance with the program performance specification.

3.2.1.1.5 Analog portion dynamic range. The dynamic range of the analog portion of the CIU shall be at least 66 dB as measured in each 110-Hz bandwidth within the 500-Hz to 3050-Hz frequency range. The dynamic range of the analog portion of the quadrature phase shifter is the ratio of the maximum signal output to the corresponding noise output level as determined by:

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20 log V/Vn

where V is the maximum voltage output without saturation and Vn is the self-noise (110 Hz bandwidth).

3.2.1.1.6 Receive analog portion response. The response of the receive analog portion shall be constant ±1 dB over the 500-Hz to 3050-Hz frequency range.

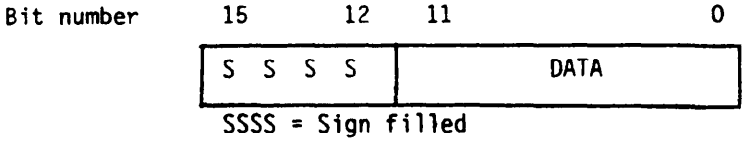
3.2.1.1.7 Receive analog portion gain. The gain of the receive analog portion shall produce a digital output between maximum digital output and 0.707 of the maximum digital output for a maximum peak analog input from the radio set.

3.2.1.2 Transmit processing. The CIU shall be capable of receiving digital signals from the AN/UYK-20 computer, processing these signals as specified herein and transferring the processed signals to the radio set. During transmission to the radio set, the audio sidetone produced by the radio set and transferred to the CIU on the audio channels (see 3.2.1.1.1) shall be received from the radio set and processed as in 3.2.1.1.

3.2.1.2.1 Output from AN/UYK-20 computer. The CIU shall receive digital inputs from the AN/UYK-20 computer on either (but not both simultaneously) of the two AN/UYK-20 computer I/O channels specified in 3.2.1.1.4. The output from the AN/UYK-20 computer shall consist of a series of 16-bit words each containing 12-bit digital values (data) on either of the two channels (only one channel shall be employed at a time). These words shall be transferred at a rate of 7040 per second.

3.2.1.2.1.1 Output data transfer format. The transfer of output data to the CIU from the AN/UYK-20 computer shall be in accordance with the Type A (NTDS slow), Category I (computer to peripheral) requirements of MIL-STD-1397.

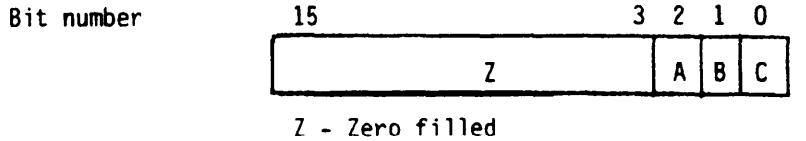
3.2.1.2.1.2 Data words from AN/UYK-20. Data words from the AN/UYK-20 computer shall be received by the CIU every 142 µs. The format of each data word shall be:



SSSS = Sign filled

When any word containing data is received from the AN/UYK-20 computer, the 12-bit value shall be applied directly (without buffer) to the CIU's D/A converter, and shall remain applied to the D/A converter until the next word is received. Storage of the current data word shall be accomplished in the CIU to conform to this requirement.

3.2.1.2.1.3 External function. The AN/UYK-20 computer shall issue external functions which shall inform the CIU whether or not to key the radio set transmitter and shall inform the CIU as to which channel shall receive the input data. Both channels shall be capable of setting the receive bits simultaneously. The format of the external function word shall be:



Z - Zero filled
A - LSB receive (1 = ON, 0 = OFF)
B - USB receive (1 = ON, 0 = OFF)
C - Transmit key (1 = ON, 0 = OFF)

The CIU shall receive the external function word from the output data lines of the AN/UYK-20 computer. The CIU shall key the radio set transmitter if bit 0 of the external function word is set. The channel that sends the latest external function word with bit 0 (transmit key) set shall be the channel from which the CIU shall accept the digital data transmitted from the AN/UYK-20 computer. The CIU shall initiate the transfer of data words from the AN/UYK-20 computer on the appropriate channel, 10 milliseconds (ms) after receiving an external function with bit 0 set.

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3.2.1.3 Digital to analog conversion. The 12-bit digital values contained in the 16-bit word from the AN/UYK-20 computer shall be digital to analog converted. The D/A converter shall be able to accept these 12-bit digital values at a rate of 7040 16-bit words per second.

3.2.1.3.1 Linearity. The D/A conversion process shall be linear to within one-half of the least significant bit at 25° C with a temperature coefficient not greater than 35 parts per million of full scale per degree C.

3.2.1.3.2 Accuracy. The D/A conversion process shall have an accuracy within ±0.3 percent of full scale over the temperature range of 0° C to 50° C.

3.2.1.3.3 Settling time. The D/A conversion process shall have a settling time of not more than 5 µs to within ±0.075 percent of full scale about its steady-state output at 25° C.

3.2.1.4 Low pass filter. The analog signal output of the D/A converter shall be applied to a low pass filter which shall smooth the signal prior to transfer to the radio set. The amplitude response of the low pass filter shall be constant within +1 dB over the 500 Hz to 3050 Hz frequency band and shall be at least 40 dB below the amplitude response in the passband (500 Hz to 3050 Hz) above 4125 Hz.

3.2.1.5 Output drivers. The analog signal output to the radio set shall be produced by the output drivers. The output drivers shall not degrade the amplitude response as a function of frequency as specified in 3.2.1.4.

3.2.1.5.1 Input to the output driver. The output of the low pass filter (see 3.2.1.4) shall, except as specified in 3.2.1.4, be applied as the input to the output drivers.

3.2.1.5.2 Output to the radio. The output driver shall produce two replicas of its input signal (the output of the lowpass filter), which shall constitute the analog signal outputs of the CIU to the USB and LSB inputs of the radio set. The signal levels of each of these two CIU outputs shall be 0 dBm ±0.5 dB into a balanced 600-ohm load when operating with the AN/UYK-20 and system program version 8.0 in net test mode. Each output transformer impedance shall be 600 ohms ±10 percent, center tapped and balanced over the frequency range of 500 to 3050 Hz.

3.2.1.6 Differential time delay. For each and every two tone combination within the 500 to 3050 Hz frequency range, the differential time delay between the two tones from the input of the low pass filter to the output of the output driver shall be less than 500 µs for the low pass filter, output driver combination.

3.2.1.7 Monitor test switch. For testing purposes, a control shall be provided to switch the USB output of the output driver to the USB input of the in-phase and phase quadrature shift section and to switch the LSB output of the output driver to the LSB input of the in-phase and phase quadrature shift section. When this control is used to switch the low pass filter output signal from the output drivers to the in-phase and phase quadrature shift section, the USB and LSB radio set receiver audio signals shall be disconnected from the in-phase and phase quadrature section and keyline signal to the radio set shall be set to OFF.

3.2.1.8 Audio monitor. The USB and LSB input signals to the in-phase and phase quadrature shift section shall also be routed from the CIU to an audio monitor unit (J-3546/USQ-74). The audio monitor unit shall provide for the monitoring of these signals with headphones. The audio monitor unit shall be suitable for mounting in the Combat Information Center (CIC). The maximum audio cable length from the CIU to the audio monitor unit shall be 182.9 meters (m) (600 feet (ft)). A type H-169 headset to be employed has a 600-ohm impedance. The audio monitor unit shall provide a jack with waterproof cover for these headphones. At the audio monitor unit, the operator shall be able to select either the USB signal, the LSB signal, or the sum of these two signals, and control the volume of the resulting signal. All circuits necessary for impedance matching, buffering and for providing a signal level to drive the headset shall be in the CIU. The headphone shall not be provided by the contractor. A minimum signal level of +2.5 dBm in either the USB or LSB position shall be provided when the input to the receive side of the CIU is 0 dBm.

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3.2.1.9 Timing and control. All timing and control for the CIU shall be generated from either an internal frequency standard or a 100-kHz external frequency standard. The operator shall be able to select either the internal or the external frequency standard by the use of a CIU front panel switch. Normal operation shall use the internal frequency standard. The contractor shall not supply the external frequency standard. The external frequency input shall be a nominal 50 ohm impedance, transformer coupled, and will accept a nominal 1 volt root-mean-square (rms) signal.

3.2.1.9.1 Internal frequency standard. The CIU internal frequency standard shall contain all the frequency determining elements necessary to generate a 1-megahertz (MHz) frequency and to meet stability requirements. The 1-MHz output shall be available at a test point at a level of at least 3 volts peak-to-peak across not less than 100,000 ohms. This test point is to be used for calibration of the crystal oscillator.

3.2.1.9.1.1 Stability. The stability of the internal frequency standard shall have the following minimum characteristics:

a. Short-term: 3 parts in 10^{11} for one second averaging time.
b. Function of power supply: Less than 5 parts in 10^{9} for a 2 percent change.
c. Function of ambient temperature: Less than 2 parts in 10^{8} over the range of 0° C to $+50^{o}$ C.

3.2.1.9.1.2 Aging. The aging rate shall not exceed 1 part in 10^{9} per 24 hours after 72 hours warm up.

3.2.1.9.1.3 Calibration control. A calibration control with a minimum adjustment range of 1 part in 10^{6} shall be provided. The internal adjustment control shall be accessible for calibration.

3.2.1.9.1.4 Rms phase deviation. The rms phase deviation of the 3520-Hz frequency (see 3.2.1.1.3) over a 13.33-ms or 22-ms period shall not be more than six degrees between any two successive periods. This frequency shall be derived from the internal or the external frequency standard.

3.2.1.9.1.5 7040-Hz clock signal. The 7040-Hz clock signal used to control the output from the AN/UYK-20 computer (see 3.2.2.1) shall also be routed to a rear panel connector. This signal shall be used to drive the external real-time clock input of the AN/UYK-20 computer. At the connector, this signal shall have a zero level of -3.0 + 0.2 VDC, a one level of 0.0 + 0.2 VDC, and a pulse duration of 71.0 µs ±5.0 µs when loaded with a resistance of between 150 and 180 ohms in series with a capacitance of between 0.0068 and 0.01 microfarad. The 7040 Hz clock signal shall be derived from the internal or the external frequency standard.

3.2.1.10 Keyline signal generation. The control circuits shall generate the keyline signal specified in 3.2.1.2.1.3. The control circuits shall turn the keyline signal to ON within 10 µs after the CIU has received an external function with bit 0 set from the AN/UYK-20 computer. The control circuits shall turn the keyline signal to OFF within 10 µs if the CIU has not received a data word from the AN/UYK-20 computer for 142 µs.

3.2.1.10.1 Keyline signal leadtime. The keyline signal voltage (ON) shall be transferred to the radio set transmitter not less than 10 ms prior to commencing the transfer of analog data signals for transmission.

3.2.1.10.2 Keyline voltage. The CIU shall supply to the radio set a keyline voltage of plus 6 (+1.0, -0.25) VDC for transmit (key ON) and 0 (+0.75, -0.25) VDC for receive (key OFF). This keyline voltage shall be capable of providing key ON voltage across a load of 3000 ohms.

3.2.1.11 Preamble transmission and reception. Behavior of the doppler tone (605 Hz) and the 2915-Hz tone during preamble transmission and reception shall conform to 3.2.1.11.1 through 3.2.1.11.4.

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3.2.1.11.1 Preamble. The preamble shall be the first five frames of every transmission and shall be a 2-tone signal consisting of a 605 Hz and a 2915 Hz tone. The 605-Hz tone shall be used for doppler correction and the 2915-Hz tone shall be used for synchronization. The 2915-Hz tone shall be phase shifted 180 degrees each successive frame enabling the receiving unit to detect frame transitions.

3.2.1.11.2 Preamble frame level. During the preamble the level of the 605-Hz tone shall be 12 ±1 dB above the level of a phase reference frame data subcarrier tone and the 2915-Hz tone shall be 6 ±1 dB above the level of a phase reference frame data subcarrier tone.

3.2.1.11.3 Phase reference frame. The phase reference frame shall immediately follow the preamble, shall be composed of the normal 16-tone composite, and shall serve as the phase reference for the first frame of the information segment.

3.2.1.11.4 Phase reference frame level. The phase reference frame shall be composed of the 16 tone composite signal. The difference in amplitude between the maximum level data subcarrier tone and the minimum level data subcarrier tone shall not exceed 1.5 dB.

NOTE: For 0 dB composite audio signal level across 600 ohms, the signal should have the following voltage levels:

Preamble:	605 Hz Doppler	= 0.692 vrms
	2915 Hz sync	= 0.346 vrms
Phase reference and information segment:	605 Hz doppler	= 0.356 volts rms
	data subcarrier tones	= 0.178 volts rms

3.2.1.12 Intermodulation distortion. Each intermodulation distortion product falling in the 500-Hz to 3050-Hz frequency band shall be a minimum of 38 dB below the level of either of any two equal amplitude data tones. In addition, for each 110-Hz frequency band with center frequencies of 935, 1045, 1155, 1265, 1375, 1485, 1595, 1705, 1815, 1925, 2035, 2145, 2255, 2365, and 2915 Hz, the peak value of the sum of all intermodulation products shall be a minimum of 40 dB below the level of two unmodulated tones at frequencies of 605 Hz and 2915 Hz. The combined power level of the two tones shall be 0 dBm ±0.5 dB, with the level of 2915 Hz tone 6 dB below the level of the 605 Hz tone. This requirement applies to all analog portions of both the receive processing circuits and the transmit processing circuits.

3.2.1.13 Bit error rate. The bit error rate (BER) performance in each sideband, USB and LSB, of the receive side of the CIU, when installed and functioning as a part of an AN/USQ-74 system, shall be not more than the BER shown in FIGURES 3 and 4. The BER requirement applies to an input signal range of from -6 to +3 dBm into 600 ohms without diversity.

3.2.2 Physical characteristics.

3.2.2.1 CV-3507/USQ-74. The case shall be designed to permit standard rack mounting in accordance with MIL-STD-189 using shelf support or hard mounting to a horizontal surface. Mounting options shall be available by means of interchangeable brackets mounted to the sides of the case. The dimensions and weight of the unit without brackets attached shall be as follows:

	Limit
Maximum (max) height	23.5 centimeters (cm) (9.25 inches (in))
Max width	43.2 cm (17 in)
Max depth	48.3 cm (19 in)
Max weight	31.8 kilograms (kg) (70 pounds (lbs))

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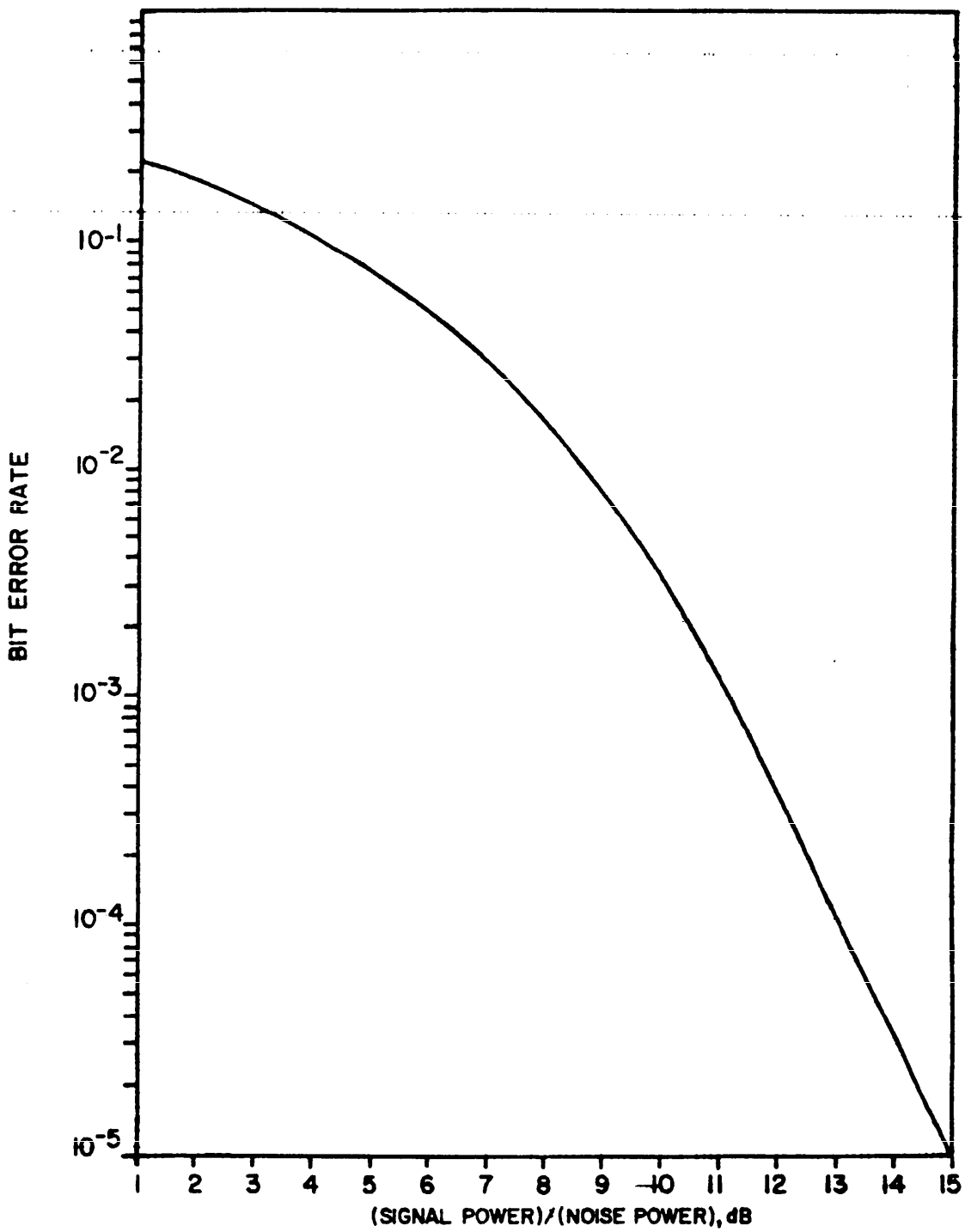


FIGURE 3. Receive side bit error rate performance. (fast rate)

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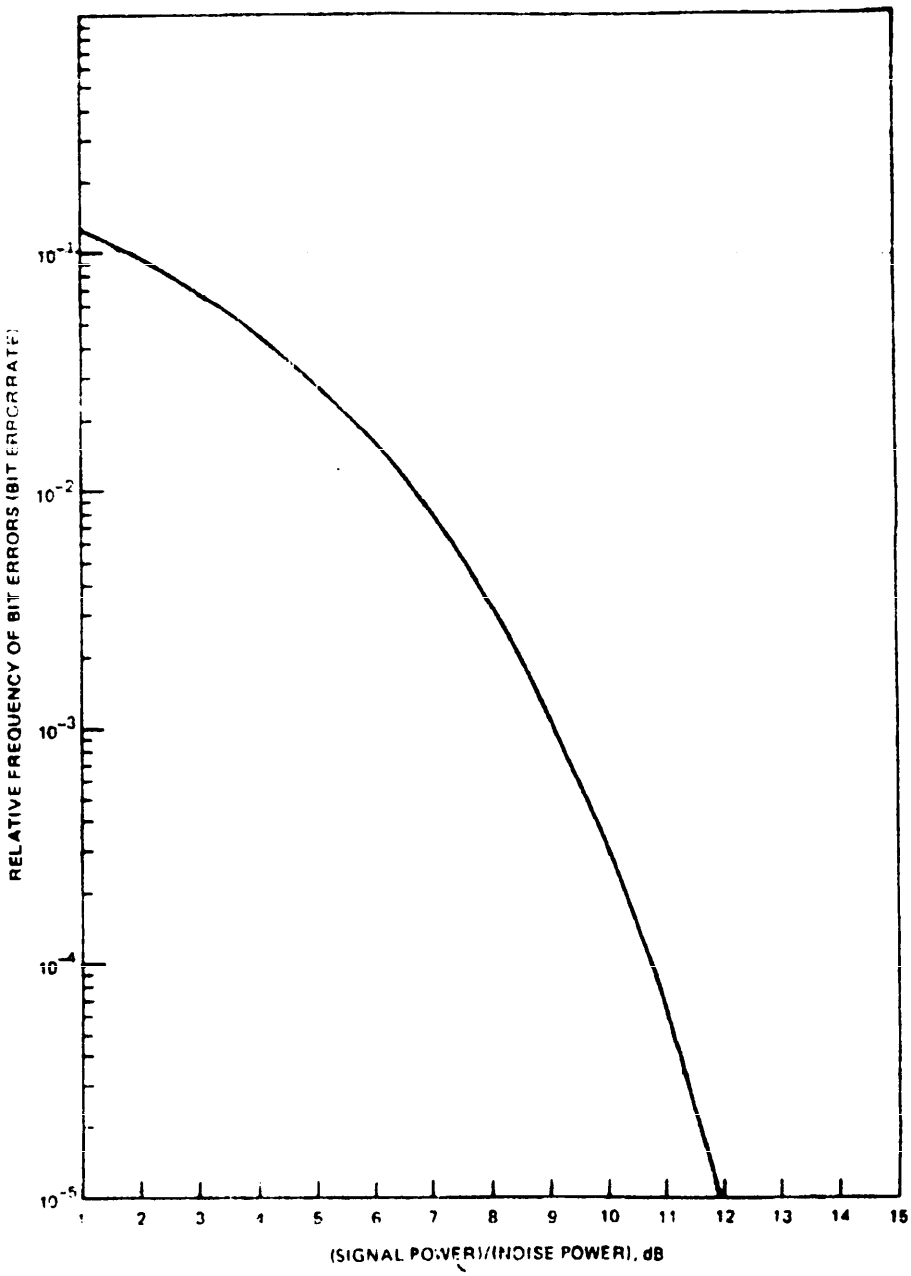


FIGURE 4. Bit error rate performance standard for slow data rate.

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When the side-mounted brackets are attached to the case and the top and bottom brackets are attached, the overall dimensions increase to the following:

Max height	26.7 cm (10.5 in)
Max width	48.3 cm (19 in)

The unit shall be delivered without brackets attached. Shelf-mounting instructions shall be included in the operation and maintenance manual.

3.2.2.2 J-3546/USQ-74. The case shall be designed to permit 8-32 bolt mounting by drilling mounting holes which coincide with the chassis mounting holes. The dimensions and weight of the audio monitor unit shall be as follows:

Width (max)	Height (max)	Depth (max)	Weight (max)
5.25 in	3.75 in	6.5 in	3 lb
13.3 cm	9.5 cm	17.5 cm	1.36 kg

3.2.3 Allen wrenches. If allen wrenches are required to remove any component of the CIU during any maintenance action, the proper allen wrenches shall be provided with each CIU. The allen wrench(es) shall have suitable stowage mounting provisions inside the case.

3.2.4 Cases. The cases shall be constructed of aluminum.

3.2.4.1 Case bonding and grounding. Case bonding and grounding shall be accomplished by preparing mounting surfaces to provide metal contact between the case and the mounting bracket, and between the mounting bracket and the mounting surface. The area around each threaded insert of each case used to secure the mounting bracket to the case shall be free of paint and chemically treated to preserve the metal and provide a ground. Mounting brackets shall be similarly treated to insure bonding and grounding.

3.2.4.1.1 Grounding stud. A .375-inch (.9525 cm) diameter grounding stud shall be installed on the rear panel of the CIU. The grounding stud shall be supplied with nut and washer.

3.2.5 Construction.

3.2.5.1 Modular construction. Modular construction shall be in accordance with MIL-E-16400 except that the standard hardware program module requirements of MIL-STD-1378 shall not apply.

3.2.5.2 Circuit boards and wire wrapping. Printed circuit boards shall be used to the maximum extent possible. Wire wrapping shall not be used.

3.2.5.3 Hardware configuration. The CIU shall be wired to accommodate all interface features and requirements as specified in 3.1. The CIU interface configurations shall be changed by the plug-in addition, removal, or replacement of appropriate hardware modules without modification to any CIU hardware. The contractor shall provide all interface modules necessary for operation with the equipment specified in 3.1.

3.2.6 Human engineering. The CIU shall conform to the human engineering criteria of MIL-STD-1472.

3.2.7 Reliability requirement.

3.2.7.1 Quantitative reliability requirement. The CIU shall have an upper test limit mean-time-between-failure (MTBF) (θ_0 as defined by MIL-STD-781) of 6000 hours. The lower test limit MTBF (θ_1 as defined by MIL-STD-781) shall be 2000 hours.

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3.2.8 Maintainability.

3.2.8.1 Quantitative maintainability requirement. The CIU shall have a mean-corrective-maintenance time (M_{ct}) not exceeding 0.25 hour and a maximum-corrective-maintenance time (95th percentile) (M_{maxct}) not exceeding 0.75 hour when corrective maintenance is accomplished at the organizational level of maintenance by the replacement of subassemblies (modules, printed circuit boards) and chassis-mounted parts (electronic, electrical, electromechanical, and mechanical). The corrective maintenance time includes localization, isolation, disassembly, interchange, reassembly, alignment, and checkout for all corrective maintenance actions. Built-in-test (BIT) circuits are included in the corrective time requirement. The equipment is considered to be in a corrective maintenance mode whenever a failure or malfunction, including a malfunction that allows degraded operation, occurs.

3.2.9 Environmental conditions.

3.2.9.1 Temperature ranges. The CIU shall conform to the temperature requirements of MIL-E-16400 for temperature Range 4.

3.2.9.2 Elevation. The CIU shall be air transportable at elevations to 15,000 meters (approximately 50,000 feet) above sea level without damage.

3.3 Parts, materials and processes. Parts, materials, and processes for the CIU shall be in accordance with the provisions of the Parts, materials and processes paragraph of MIL-E-16400 except as otherwise specified herein.

3.3.1 Parts control. Procedures for selection of electronic, electrical, electromechanical and mechanical parts shall be in accordance with MIL-STD-965, Procedure I.

3.3.1.1 Nonstandard parts approval. All nonstandard parts shall have been approved prior to final inspection and tests.

3.3.2 Choice of parts and materials. The performance of the equipment shall not be dependent on the individual selection of parts unless each selection is specifically approved by the procuring activity. If such individual parts selection is approved, sufficient information and data shall be included as is necessary in all applicable technical manuals and engineering drawings to enable designated repair activities to perform such selection.

3.3.3 Interchangeability. All modules, assemblies, subassemblies, and other elements and components designed to perform the same function and having a common design in different converter interface units together with replacement components for the CIU shall be interchangeable without drilling, filing, or undue use of force. The replacement component, when installed, shall not result in departure from the performance specified herein.

3.3.3.1 Interchangeability with Government furnished equipment. The CIU produced to this specification shall be physically, electrically, and functionally interchangeable with the GFE. Like units, modules, PC boards, and assemblies shall be physically, electrically, and functionally interchangeable between the CIU produced to this specification and the PC boards and power supplies from one CIU produced to this specification shall be interchanged with the Government-furnished CV-3507/MSQ-74 and a bit error test (4.5.4.17) conducted on each unit. Individual items shall not be hand-picked for fit or performance. Interchangeability requirements shall be two-way.

3.3.4 Cabling and connectors. Cables and the associated multicontact connectors used in interconnecting major assemblies shall have a minimum of 10 percent spare unused contacts and associated connectors, but in no case shall there be less than two spares. Multicontact connectors used for primary power shall have one spare contact. The mating connectors with backshells shall be supplied with the CIU.

3.3.4.1 Connectors. Electrical interface connectors shall be in accordance with MIL-C-5015, except the 100 kilohertz (kHz) external frequency standard input connector shall be a BNC type in accordance with MIL-C-39012. The connectors for the external frequency standard input and the keyline output to the radio set shall have captive protective covers.

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3.3.5 Semiconductor devices. Semiconductor devices composed of germanium shall not be used. Plastic encapsulated semiconductor devices shall not be used. Standard discrete semiconductor parts shall be selected from those listed in MIL-STD-701.

3.3.6 Carcinogens. Materials identified as carcinogens (cancer producing substances) in 29 Code of Federal Regulations, Part 1910, Subpart Z-Toxic and Hazardous Substances, shall not be used as part of the equipment.

3.3.7 Material toxicity. No materials shall be used in the equipment which, when exposed to a temperature range of -65° Celsius (C) to +125° C, emit air contaminants in excess of the threshold values specified in 29 Code of Federal Regulations, Part 1910, Subpart Z-Toxic and hazardous Substances.

3.3.8 Fungus. The use of fungus inert and fungus nutrient materials shall conform with Requirement 4 of MIL-STD-454.

3.3.9 Government furnished material. Government furnished material (GFM), if any, shall be as specified (see 6.2).

3.4 Electrical design and construction.

3.4.1 Power requirements. The CIU shall operate from a single-phase 115 volts alternating current (VAC) +10 percent, 50, 60 or 400 Hertz (Hz), each ±5 percent, power source. Input power to the CIU shall not exceed 60 watts. The CIU shall perform in accordance with the Electrical power characteristics paragraph of MIL-E-16400.

3.4.1.1 Hum. Power supply ripple at 47 Hz shall be less than 2 percent and shall improve with an increase in line frequency.

3.4.2 Power interface. The CIU shall be designed to operate with the electrical power interface characteristics specified for Type 1 systems of Section 103 of MIL-STD-1399 except as specified in 3.4.1. With the exception of the 700-volt transient test, the system shall be designed to withstand the electrical tests specified in MIL-E-16400. The insulation resistance test, corona measurement test, and dielectric withstanding voltage tests are not required.

3.4.3 Electrical overload protection. Fuses shall be used to provide overload protection in accordance with MIL-E-16400. Both primary power lines shall be fused. Fuses shall be mounted on the equipment front panel in blown fuse indicating fuseholders which conform to MIL-E-19207.

3.4.4 Electromagnetic compatibility (EMC). The CIU shall be self-compatible and shall meet the requirements of MIL-STD-461 for Class A4 equipments. MIL-HDBK-241 may be used for guidance in electromagnetic interference (EMI) reduction in power supply design.

3.4.4.1 Electromagnetic pulse (EMP). The CIU interface connections shall withstand the following maximum common-mode current and voltage transients at each non-antenna interface pin:

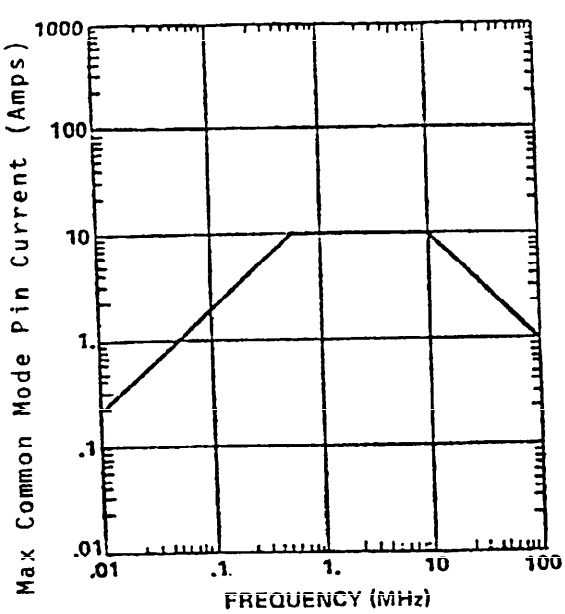
$Ipin\ (t) = \pm\ Ip(fo)\ e^{-\pi fot/Q} \sin\ (2\ \pi fot)$ and
$Vpin\ (t) = \mp\ Vp(fo) e^{-\pi fot/Q} \sin\ (2\ \pi fot)$, where
fo = resonant frequency in Hertz
t = time in seconds
Q = decay factor = 16 ± 4 and Ip(fo) and Vp(fo) are given as a function of the resonant frequency in FIGURE 5.

3.5 Maintenance design.

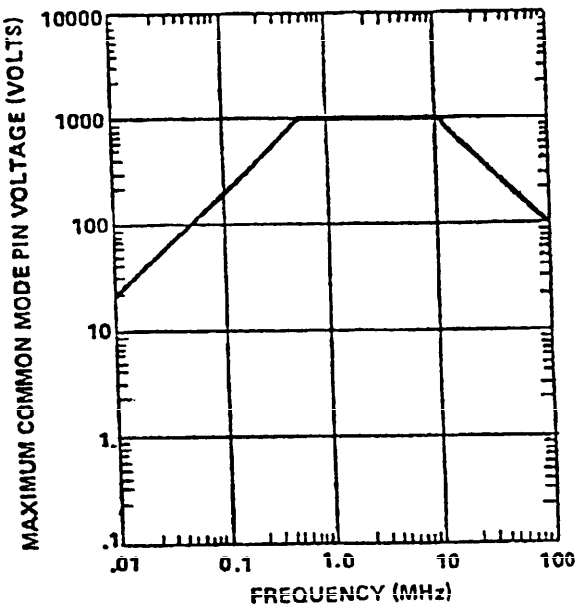
3.5.1 Test measurement and diagnostic equipment (TMDE).

3.5.1.1 Test provisions. The CIU shall contain the test provisions specified in 3.5.1.1.1 and 3.5.1.1.2.

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INTERFACE PIN CURRENT FOR EQUIPMENT



INTERFACE PIN VOLTAGE FOR EQUIPMENT

FIGURE 5. Resonant frequency function of Ip (fo) and Vp (fo).

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3.5.1.1.1 Class A test provisions. Class A test provisions shall provide a means to verify the on-line equipment is operating properly. Class A provisions shall be accomplished by the use of the test provisions of the Government furnished AN/USQ-74 software for the AN/UYK-20 computer with the CIU connected to the AN/UYK-20 computer and compatible LINK 11 radio (see 3.1.2.1). Under the Class A test provisions the contractor shall only be responsible for the CIU being compatible with maintenance tests three and four of the Government furnished AN/USQ-74 software as defined in NOSC TM 344 and 348, both with Change 1 of June 1979.

3.5.1.1.2 Class D test provisions. Class D test provisions shall provide a means to fault isolate electrical functions (for example, integrated circuits, filter, A/D, and D/A). Class D provisions may be accomplished by the use of test points in conjunction with the use of automatic test equipment (ATE) or general purpose electronic test equipment (GPETE) and shall have the capability of isolating all the failures to the electrical functional items on a subassembly. Neither ATE nor GPETE shall be supplied by the contractor.

3.5.1.2 Test point criteria. Test points shall satisfy the requirements specified in 3.5.1.2.1, 3.5.1.2.2 and 3.5.1.2.3.

3.5.1.2.1 Signal quality and circuit performance. Test points, test jacks, or both, shall be provided to permit the injection of signals and the monitoring of signals at the input and output terminals of the subassembly.

3.5.1.2.2 Fault isolation. Test points, test jacks, or both shall be provided to permit the injection and measurement of signals at input and output terminals of the subassemblies for the purpose of isolating all faults to the electrical functions (for example, integrated circuits, filter, A/D, and D/A).

3.5.1.2.2.1 Board or cable extenders. Board extenders or cable extenders, or both, shall be provided to enable fault isolation on the subassemblies with the subassemblies removed from the CIU. Test points shall be accessible with the subassemblies in place.

3.5.1.2.3 Required test points. Test points are required for but not limited to the following signals and locations:

a. Signal ground on each subassembly for convenience of making signal/voltage measurements.
b. Each input to the in-phase and phase quadrature shift section.
c. Each output of the in-phase and phase quadrature shift section.
d. Each output of the sampling process.
e. The multiplexer output which is not connected to the A/D converter, if a two output analog multiplexer is used in the sampling and A/D conversion process.
f. The least significant bit of the output of the A/D conversion process.
g. If operational amplifiers are used in the in-phase and phase quadrature shift section, between each stage of the in-phase and phase quadrature shift section.
h. The output of the D/A conversion process.
i. The output of the low pass filter.
j. Each output of the output drivers.
k. The keyline.
l. The input for the external frequency standard.
m. The output(s) of the internal frequency standard.
n. Each stage of the process used to convert the 100 kHz signal provided by the external frequency standard to the timing signals required by the CIU.
o. Each control signal associated with input/output (I/O) operations with the AN/UYK-20 computer.
p. Power supply (each DC power supply level).

3.5.1.3 GPETE. GPETE to be used at all levels of maintenance shall be selected from the standard and substitute standard test equipments listed in MIL-STD-1364.

3.6 Mechanical design and construction.

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3.6.1 Degree of enclosure. The CIU enclosure and the J-3546/USQ-74 enclosure shall be dripproof (15 degrees) in accordance with MIL-STD-108.

3.6.2 Magnetic characteristics. The CIU shall employ no material which has a relative magnetic permeability greater than 2.0, except for magnetic materials absolutely required for the electrical functioning of the equipment.

3.6.3 Controls, indicators, and panel layouts. The controls and indicators, and their arrangement on operator and maintainer panels, shall conform to the criteria of MIL-STD-1472 and shall be in accordance with the Controls, indicators and panel layout paragraph of MIL-E-16400, except that color coding and marking of pin jacks and other test points shall be in accordance with MIL-STD-415. The abbreviations of MIL-STD-12 shall be used for equipment labeling where applicable.

3.6.3.1 Controls and displays. The following controls and displays shall be provided on the front panel of the CIU:

a. A main AC input power switch and an associated AC power ON indicator light (green).
b. The monitor test function switch described in 3.2.1.7.
c. An indicator light (green) to indicate when the keyline is active.
d. A switch for selection of either the external or the internal frequency standard
e. A time meter in accordance with MIL-M-7793 which shall indicate the elapsed time of operation. The range of the meter shall be 0 to 9999 hours and shall repeat. The meter shall not be capable of being reset. The elapsed time meter may be mounted inside the unit in a location that is visible when the front panel is lowered.

The following controls shall be provided on the front panel of the audio monitor unit, J-3546/USQ-74 jackbox:

f. A switch for the selection of the USB signal, the LSB signal, or the sum of these two signals.
g. A volume control.

3.7 Thermal design. Automatic shutdown capability is not required. The CIU shall be convection air cooled without case or panel vents.

3.8 Nameplates and product marking. Nameplates and product markings shall be prepared in accordance with MIL-P-15024. No identification of the contractor other than that allowed for the nameplate shall appear on the exterior of the system. Markings on the CIU shall be in accordance with MIL-STD-130 and shall include serialization of the unit, replaceable assemblies, subassemblies, and printed circuit (PC) boards.

3.9 Safety. Safety requirements shall be as specified in the Safety criteria paragraphs of MIL-E-16400, except as specified herein. The CIU shall be designed so that no interlocks are required in accordance with Requirement 1 of MIL-STD-454.

3.9.1 Leakage current. The leakage current of the equipment shall not exceed 5 milliamperes (mA) alternating current (AC) or direct current (DC).

3.9.2 Equipment power connections. Equipment power connections for 115 volts alternating current (VAC) input power and interconnecting AC power cable connectors between equipments shall have only four pins, designated A, B, C and D. Pin assignments and color codes shall be as specified in TABLE I.

TABLE I. Power connection pin assignments and color codes.

Pin	Conductor assignment	Color code
A	115 VAC return	White
B	Safety ground	Green
C	115 VAC high (hot)	Black
D	Spare	None

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Black and white color code shall be maintained from the input connector pins to all components having the same voltage and frequency as the input power.

3.9.3 Electrical power on/off switch. A switch for disconnecting the CIU from all electrical power systems shall be mounted on the equipment front panel and its function clearly labeled. The switch shall break all power conductors (except the safety ground conductor) of the power circuit.

3.9.3.1 Power indicator lamp. A green power indicator lamp shall be mounted near the switch to indicate when the CIU is energized.

3.10 Workmanship.

3.10.1 General workmanship. Workmanship shall conform to Requirement 9 of MIL-STD-454.

3.10.2 Workmanship screen. All equipment shall withstand a defect detection vibration screen of random type vibration at $0.04g^2$/Hz ±3 dB from 80 to 350 Hz 3 dB per octave roll off on each end of frequency range and a defect detection temperature cycling screen.

3.11 Personnel qualifications. The CIU shall be capable of being operated and maintained by a data systems technician military grade E-4 as classified in NAVPERS 18068.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Government verification. All quality assurance operations performed by the contractor will be subject to Government verification at any time. Verification will consist of, but is not limited to, a) Surveillance of the operations to determine that practices, methods, and procedures of the written quality program are being properly applied, b) Government product inspection to measure quality of the product to be offered for acceptance, c) Government inspection of delivered products to assure compliance with all inspection requirements of this specification. Failure of the contractor to promptly correct deficiencies discovered by him or of which he is notified shall be cause for suspension of acceptance until corrective action has been taken or until conformance of the product to prescribed criteria has been demonstrated.

4.1.2 Quality assurance terms and definitions. Quality assurance terms used in this specification shall be as defined in MIL-STD-109.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3)
b. Quality conformance inspection (see 4.4)
    1. Production inspection (see 4.4.1)
    2. Production control inspection (see 4.4.2)
    3. Environmental inspection (see 4.4.3)
c. Inspection of preparation for delivery (see 4.8)

4.3 First article testing. Unless otherwise specified (see 6.2), one unit shall be required for first article inspection. First article inspection shall consist of all examination and testing necessary to determine compliance with the requirements of this specification. First article inspection shall include the tests specified in TABLE II.

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TABLE II. Examinations and tests.

Subject	Requirement	Test method	First article	Quality conformance inspection		
				Group (A)	Group (B)	Group (C)
Surface examination	3.2.2	4.5.2	X	X		
Weight	3.2.2	4.5.2	X	X		
Size	3.2.2	4.5.2	X	X		
Parts and materials	3.3	4.5.2	X	X		
Finish	3.1	4.5.2	X	X		
Marking	3.8	4.5.2	X	X		
Safety	3.9	4.5.2	X	X		
Workmanship	3.10	4.5.3	X	X		
Temperature sequence						
Low temperature	3.2.9.1	4.5.5.1	X			
High temperature	3.2.9.1	4.5.5.1	X			
Humidity	3.1	4.5.5.2	X			
Fungus	3.3.8	4.5.5.4	X			
Workmanship screen						
Pre-performance	3.10.2	4.5.6.1		X		
Post-performance	3.10.2	4.5.6.2		X		
Vibration	3.10.2	4.5.6.3		X		
Temperature cycling	3.10.2	4.5.6.4		X		
Human engineering	3.2.6	4.5.9	X			
Shock	3.1	4.5.5.3	X			
Vibration	3.1	4.5.1	X			
Enclosure	3.6.1	4.5.10	X			
Magnetic	3.6.2	4.5.1	X		X	
Steady-state voltage and frequency	3.4.1	4.5.7.1	X		X	
Transient voltage	3.4.1	4.5.1	X			
Transient frequency	3.4.1	4.5.1	X			X
Spike voltage	3.4.1	4.5.7.2	X		X	
Power interruption	3.4.1	4.5.1	X			X
Power and power factor	3.4.1	4.5.1	X			
Interchangeability and compatibility	3.3.3, 3.3.3.1	4.5.4.17	X			X
Leakage current	3.9.1	4.5.7.4	X			
Power supply hum	3.4.1.1	4.5.7.3	X			
Electromagnetic compatibility	3.4.4, 3.4.4.1	4.5.8.1, 4.5.8.2	X		X	
Thermal design	3.7	4.5.1	X			
Elevation	3.2.9.2	4.5.11	X			X
Reliability	3.2.7.1	4.6	X			
Maintainability	3.2.8.1	4.7	X			
Performance tests	-	4.5.4				
Sampling and A/D conversions	3.2.1.1.3	4.5.4.3	X	X		
Digital to analog conversion	3.2.1.3	4.5.4.8	X	X		
Timing and control	3.2.1.9	4.5.4.14	X	X		
Internal frequency standard	3.2.1.9.1	4.5.4.14	X	X		
Intermodulation distortion	3.2.1.12	4.5.4.15	X		X	
Quadrature phase shift	3.2.1.1.2	4.5.4.2	X		X	

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TABLE II. Examinations and tests. (Continued)

Subject	Requirement	Test method	First article	Quality conformance inspection		
				Group (A)	Group (B)	Group (C)
Analog portion dynamic range	3.2.1.1.5	4.5.4.6	X		X	
Receive analog portion gain/response	3.2.1.1.6, 3.2.1.1.7	4.5.4.7	X		X	
~~Low pass filter~~	~~3.2.1.4~~	~~4.5.4.9~~	~~X~~		~~X~~	
Differential time delay	3.2.1.6	4.5.4.11	X		X	
Rms phase deviation	3.2.1.9.1.4	4.5.4.14	X		X	
Receive input	3.2.1.1.1	4.5.4.1	X	X		
Input to AN/UYK-20 computer	3.2.1.1.4	4.5.4.4	X	X		
Output from AN/UYK-20 computer	3.2.1.2.1	4.5.4.4	X	X		
Output drivers	3.2.1.5	4.5.4.10	X	X		
Monitor test switch	3.2.1.7	4.5.4.12	X	X		
Audio monitor	3.2.1.8	4.5.4.13	X	X		
Keyline signal generation	3.2.1.10	4.5.4.5	X	X		
Keyline voltage	3.2.1.10.2	4.5.4.5				
7040 Hz clock signal	3.2.1.9.1.5	4.5.4.14.1	X	X		
Controls and display	3.6.3.1	4.5.4.16	X	X		
Operation w/ GFE:	3.1.3	-				
Bit error test w/ noise	3.2.1.13	4.5.4.18	X		X	
Bit error test w/o noise	3.2.1.13	4.5.4.17	X	X		

4.4 Quality conformance inspection.

4.4.1 Production inspection (Group A). Production inspection shall be conducted on each equipment offered for delivery. This inspection shall comprise such examination and tests which will prove the workmanship and reveal omissions and errors of the production process such as, functional and performance tests at a limited number of points in the required range, tests which detect deviations from design, tests of controls and adjustments, and tests which detect hidden defects of materials. Production inspection shall include the examination and tests specified in TABLE II (Group A).

4.4.2 Production control inspection (Group B). Production control inspection shall be conducted on a sampling basis as specified in 4.4.2.1. This inspection shall consist of the examinations and tests which encompass functional and performance tests throughout the entire range of operation; tests which will detect any deterioration of the design by wear of such items as dies, molds, and jigs, and by substitution of different parts; tests which detect deviations in the processing of the materials; tests to determine temperature rise produced in operation and ability of equipment to withstand this heat; tests of efficiency; and tests of the performance with other equipment in a system. These tests shall be performed on the complete CIU as offered for delivery. Where necessary, Group B tests may be performed at module or printed circuit board level. Production control inspection shall include the examinations and tests specified in TABLE II (Group B).

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4.4.2.1 Sampling plan for Group B inspection. Equipment submitted for Group B sampling tests shall be selected by the contractor under the supervision of the Quality Assurance Representative and shall be representative of current production. Group B inspection shall be performed on CIUs that have passed Group A inspection. One CIU shall be selected from each successive 15 CIUs, or fraction thereof, produced.

4.4.3 Environmental inspection (Group C). Environmental inspection shall be conducted on a sampling basis as specified in 4.4.3.1. This inspection shall encompass environmental tests to prove the durability of the materials, parts, units, and the equipment as a whole; life tests, simulated service tests; test of the effects of changes of environment (such as, extremes of temperature and humidity, effect of salt air); and tests of the effects of shock, vibration, and inclination. Environmental inspection shall include the examination and tests specified in TABLE II (Group C).

4.4.3.1 Sampling for environmental inspection. One equipment shall be selected from each successive 40 CIUs, or fraction thereof, produced. The first sample CIU shall be selected from the first months production.

4.4.3.2 Nonconforming environmental sample units. If a sample unit fails the inspection specified in 4.4.3, the contractor shall immediately investigate the cause of failure and shall report to the Quality Assurance Representative the results thereof and details of the corrective action taken to correct units of product which were manufactured under the same conditions, with the same materials, processes, and so forth. The CIU shall then be retested.

4.4.4 Reinspection of conforming production control and environmental test units. Unless otherwise specified in the contract (see 6.2), sample units which have been subjected to and have passed production control or environmental inspections, or both, may be accepted on the contract provided they are resubjected to and pass production inspection after repair of all visible damage.

4.4.5 Test equipment and test conditions. Test equipment and test conditions for first article inspection and quality conformance inspection shall conform to the following quality assurance provisions of MIL-E-16400:

a. Test facilities, chambers, and apparatus
b. Standard test conditions
c. Performance of test

4.5 Test methods.

4.5.1 Examinations and tests. Unless otherwise specified herein, the inspections specified in TABLE II shall be performed in accordance with the test methods specified in MIL-E-16400.

4.5.2 Surface examination. The equipment shall be examined to ensure compliance with the applicable referenced specifications with respect to:

(a) Weight (see 3.2.2.1)
(b) Size (see 3.2.2.1)
(c) Parts and materials (see 3.3)
(d) Finish (see 3.1)
(e) Marking (see 3.8)
(f) Safety (see 3.9)

In the process of examination, the item shall not be disassembled in a manner that will affect the performance, durability or appearance of the item. The examination shall include a check of all operating controls, circuit functions, test provisions and adjustments.

4.5.3 General workmanship. The equipment, including subassemblies and assemblies, shall be examined for workmanship and soldering during the fabrication and assembly process and at the end item level for conformance to the requirement of 3.10. Each solder connection and its associated wiring or leads shall be visually examined.

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4.5.4 Performance test. The equipment shall be subjected to a performance test including all operating controls to determine compliance with the performance parameters and safety requirements herein. The performance test shall include the tests of 4.5.4.1 through 4.5.4.19.

4.5.4.1 Input test. Sinusoidal signals of from six volts peak-to-peak to 12 volts peak-to-peak from a 600 ohm source shall be applied to each of the two audio inputs. The test frequencies shall be 500 Hz, 1 kHz, 2 kHz, and 3 kHz. If no saturation of any circuit in the receive processing section up to the sampling and A/D converter section occurs, the CIU shall have passed the test.

4.5.4.2 Quadrature phase shifting test. Sinusoidal signals of the frequencies in 4.5.4.6 shall be applied to each quadrature phase shift circuit. For each sinusoidal signal applied:

a. The phase difference between the LSB in-phase output signal and the USB in-phase output signal shall be measured.
b. The phase difference between the LSB in-phase output signal and the LSB quadrature phase output signal shall be measured.
c. The phase difference between the USB in-phase signals and the USB quadrature phase output signal shall be measured.
d. The amplitude response of all four output signals shall be measured.

4.5.4.3 Sampling and A/D conversion test. The sampling rate of 3520 samples per second per each of the four signal outputs of the phase quadrature circuits shall be verified. The aperture width, the linearity of the A/D conversion processes, and the accuracy of the A/D conversion processes shall be measured and verified as to conformance with requirements over the full input voltage range. The 12-bit binary range and two's complement output of the A/D conversion shall be verified by monitoring of the two CIU outputs to the AN/UYK-20 computer.

4.5.4.4 I/O to AN/UYK-20 computer test. The signal timing, levels, and sequence of operation of the two I/O channels and the circuits controlling these channels shall be measured and verified as compatible with the AN/UYK-20 computer and in accordance with MIL-STD-1397. The inputs to the computer shall be monitored to verify the format of the input data and the interrupt word. The input of the interrupt word shall be verified if the AN/UYK-20 computer does not accept an input word within 71 s after the word is available. The outputs from the AN/UYK-20 computer shall be monitored to verify the CIU will accept words at a rate of 7040 words per second (one word every 142 s) and with the format specified. All possible external functions shall be sent by the AN/UYK-20 on each channel, and the response on the input data channels to the AN/UYK-20 from the CIU shall be monitored to verify proper responses. This I/O test shall be performed by the Government as part of the first article test. This test will be required at the discretion of the Government when configuration changes occur.

4.5.4.5 Keyline voltage test. The keyline voltage shall be measured to verify it is +6 (+1.0, -0.25) volts DC for transmit, key ON, and 0 (+.75, -0.25) volts DC for receive, key OFF. Measurements shall be made across a 3000-ohm resistive load.

4.5.4.6 Receive processing, analog portion dynamic range test. The dynamic range of each channel shall be determined by measuring the levels of sinusoidal signals required to saturate the receive channels of the CIU, and the level of the self-noise of these receive channels as a function of bandwidth. These sinusoidal signals shall be applied at each of the audio inputs to the CIU and shall be tested at 500, 605, 935, 1045, 1155, 1265, 1375, 1485, 1595, 1705, 1815, 1925, 2035, 2145, 2255, 2365, 2915, and 3050 Hz. The dynamic range shall be the smallest difference between the levels of the sinusoidal signals and the self-noise.

4.5.4.7 Gain test. The gain of the receive processing channels shall be measured and verified to be constant ±1 dB. An input signal of 12 volts peak-to-peak shall be applied to each input to the CIU, and the digital output examined to verify that this output is between maximum and 0.707 maximum. Gain shall be measured at the frequencies listed in 4.5.4.6.

4.5.4.8 D/A conversion test. The linearity, accuracy, and settling time of the D/A converter shall be measured and verified as to conformance with requirements. The D/A converter output shall be monitored to verify that it is accepting 12-bit digital values at a rate of 7040 per second and converting these values to analog form in accordance with the specification.

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4.5.4.9 Low pass filter and output driver test. The amplitude response of the low pass filter and output driver shall be measured over the frequency range of 500 Hz to 14,000 Hz to verify that the amplitude response is constant ±1 dB in the 500 Hz to 3050 Hz passband and at least 40 dB below the passband amplitude response above 4125 Hz.

4.5.4.10 Output driver test. The two signal outputs from the CIU to the radio set shall be terminated in 600-ohm balanced loads and measured to determine if the signals levels are 0 dBm +0.5 dB. This test shall be performed with the CIU connected to the AN/UYK-20 computer and with the CIU receiving data from the AN/UYK-20. The AN/UYK-20 shall be operating with Government-furnished approved program software version 8.0 operating in net test mode. Later versions of the AN/USQ-74 program software may be used at contractor discretion.

4.5.4.11 Differential time delay test. Sinusoidal signals at frequencies of 500, 605, 935, 1045, 1155, 1265, 1375, 1485, 1595, 1705, 1815, 1925, 2035, 2145, 2255, 2365, 2915, and 3050 Hz shall be applied to the input of the low pass filter. The differential time delay of each of these sinusoidal signals relative to the 2915 Hz signal shall be measured at both outputs of the output driver.

4.5.4.12 Monitor test function test. The operation of monitor test function switch shall be activated to determine if it does switch the USB output of the output driver to the USB input of the phase quadrature section and to switch the LSB output of the output driver to the LSB input of the phase quadrature section, and does disconnect the inputs from the radio set.

4.5.4.13 Audio monitor test. The audio monitor unit shall be checked with a set of 600 ohms impedance headphones to determine that aural monitoring of the USB signal to the phase quadrature section, the LSB signal to the phase quadrature section, and the sum of these two signals is possible. The volume control shall also be tested. This test shall be conducted with the maximum cable length of 182.9 m (600 ft) between the audio monitor unit and the CIU. The output headphone signal shall be measured in accordance with 3.2.1.8.

4.5.4.14 Timing and control test. The time and phase stability of the internal frequency standard shall be measured to verify that the frequency standard conforms to the requirements of 3.2.1.9. External 100-kHz signal shall be applied to the external standard input connection and the clock signal verified for proper frequency output.

4.5.4.14.1 7040 Hz clock signal test. The rear panel connector for the 7040 Hz clock signal shall be loaded with a resistance of between 150 and 180 ohms in series with a capacitance of between 0.0068 and 0.01 microfarad. With the CIU operating, an oscilloscope shall be connected across this load. The oscilloscope display shall be observed to determine the zero level voltage, the one level voltage, and the pulse duration. The 7040-Hz signal shall be measured to determine if the correct frequency has been derived.

4.5.4.14.2 3520 Hz signal. The signal shall be measured to verify that the frequency conforms to the requirement of 3.2.1.9.1.4.

4.5.4.15 Intermodulation distortion test. Intermodulation distortion shall be measured in both the receive processing circuits and the transmit processing circuits. The audio frequency input shall consist of two unmodulated tones, one at a frequency of 605 Hz, and one at a frequency of 2915 Hz. The 605-Hz tone shall be the reference tone and the 2915-Hz tone shall have an amplitude of 6 dB below the reference tone. The power level of the combined two tones shall be 0 dBm +0.5 dB. The measurement shall be performed by applying the audio frequency tones to the circuits to be tested and, with an audio frequency spectrum analyzer having a resolution of 10 Hz or less and a dynamic range of 50 dB or greater measuring the maximum amplitude of all intermodulation products in the 500-Hz to 3050-Hz frequency range. These amplitudes shall be examined to determine that each intermodulation product is a minimum of 30 dB below the reference tone. In addition, these amplitudes shall be examined to determine that the peak value of the sum of all intermodulation products is a minimum of 30 dB below the reference tone for each 110 Hz frequency band, with center frequencies of 935, 1045, 1155, 1265, 1375, 1485, 1595, 1705, 1815, 1925, 2035, 2145, 2255, and 2365 Hz.

4.5.4.16 Controls and display. All control functions and display shall be verified for proper operation in accordance with 3.6.3.1, and 3.9.3.

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4.5.4.17 Bit error test and operation with GFE without noise. A one-hour bit error rate (BER) test shall be established using the AN/UYK-20 computer and Government-furnished AN/USQ-74 approved program software Version 8.0, except later versions of the AN/USQ-74 program software may be used at contractor discretion. The one hour BER test shall be conducted with the CIU's monitor test switch in the operate position, loop back, the CIU energized, the CIU connected to the AN/UYK-20 computer, and the computer running a net test program for bit error analysis. The data to be collected for this test shall be the number of bit errors in the data returned to the AN/UYK-20 computer by the CIU in one hour. The Government furnished-software is capable of counting the number of bit errors in one hour. This shall be an error-free test.

4.5.4.18 Bit error test with noise. BER tests shall be established to measure bit errors when signal and noise are introduced into the receive input channels. Separately, each channel, LSB and USB shall be tested at fast and slow data rates. The tests shall be made with input ~~signal levels of +3 dBm, 0 dBm, and -6 dBm. The added white noise shall be at levels of -3 dBm,~~ -6 dBm, -10 dBm, and -15 dBm for fast data rates and -3 dBm, -6 dBm, -10 dBm, and -12 dBm for slow data rates. The BER shall not exceed the limits shown in FIGURE 3 for fast data rates and FIGURE 4 for slow data rates.

4.5.5 Environmental tests.

4.5.5.1 Temperature test. The temperature test shall be conducted in accordance with the temperature test of MIL-E-16400. The pretest data shall be that data collected from performing the tests which are required in compliance with the production testing. Pretest data includes data from a one-hour BER test (see 4.5.4.17). The AN/UYK-20 computer shall be outside the test chamber. The performance check during the temperature test shall be a one-hour BER test using the AN/UYK-20 computer and the net test of 4.5.4.17. The post-test data shall consist of data resulting from a repeat of all tests and inspections used to obtain the pretest data, including the one-hour BER test; however, post testing shall be done only once for both temperature and humidity tests.

4.5.5.2 Humidity test. The humidity test shall be conducted in accordance with the humidity test of MIL-E-16400. The pretest data shall be from a one-hour BER test (see 4.5.4.17). The AN/UYK-20 shall be outside the temperature humidity chamber. The performance check during the humidity test shall be a one-hour BER test using the AN/UYK-20 computer and net test of 4.5.4.17. The post-test data shall consist of data resulting from a repeat of all tests and inspectors that were used to obtain the pretest data for the temperature test of 4.5.5.1, including the one-hour error rate test. The post testing shall be done only once for both temperature and humidity tests.

4.5.5.3 Shock test. The shock test shall be conducted in accordance with MIL-S-901. The equipment under test shall be mounted on standard mounting fixtures in any mode of operation and with simulation of external loads (cables and so forth). At the end of the shock test, a bit error test (4.5.4.17) shall be performed.

4.5.5.4 Fungus. The contractor shall certify by analysis conformance to Requirement 4 of MIL-STD-454 (see 3.3.8).

4.5.6 Workmanship screen.

4.5.6.1 Pre-performance tests. (a). Prior to the workmanship screen, each CIU shall be tested and (if applicable) adjusted for the correct power supply voltages, the output signal to the radio, the keyline voltage, the 7040 Hz clock signal and the functions of all controls and displays.

(b). At room temperature, a 15-minute BER test shall be established using the AN/UYK-20 computer and Government-furnished AN/USQ-74 approved program software Version 8.0, except later versions of the AN/USQ-74 program software may be used at contractor discretion. The one-hour BER test shall be conducted with the CIU's monitor test switch the operate position, the CIU energized, the CIU connected to the AN/UYK-20 computer and the computer running a net test program for bit error analysis. The data to be collected for this test shall be the number of bit errors in the data returned to the AN/UYK-20 computer by the CIU. The Government-furnished software is capable of counting the number of bit errors in one hour.

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4.5.6.2 Post-performance tests. Upon successful completion of workmanship screening (vibration, temperature and voltage cycling), a 15-minute BER test (see 4.5.6.1.b) shall be performed including a retest of the keyline voltage, the 7040 Hz clock signal, the output voltages, and the functions of all controls and displays.

4.5.6.3 Vibration.

4.5.6.3.1 Prior to conducting temperature cycling, (4.5.6.4) vibration shall be performed on each CIU. The vibration may be performed at the module, drawer, or end item level. All the hardware including cables and connectors shall be exposed to vibration.

4.5.6.3.2 The vibration shall be random, or subject to Procuring activity approval, pseudorandum or complex wave-form vibration, for an accumulated time of ten minutes in the axis deemed most susceptible to vibration excitation. ~~All test items shall be hard mounted (without~~ shock isolators) and subjected to the vibration conditions of FIGURE 7. The control accelerometer shall be located next to one of the mounting points of the item under test. Equipment having a bandwidth no greater than 10 Hz for vibration frequencies up to 500 Hz and 100 Hz for vibration frequencies above 500 hz shall be used for the control and analysis of the acceleration spectral density (ASD).

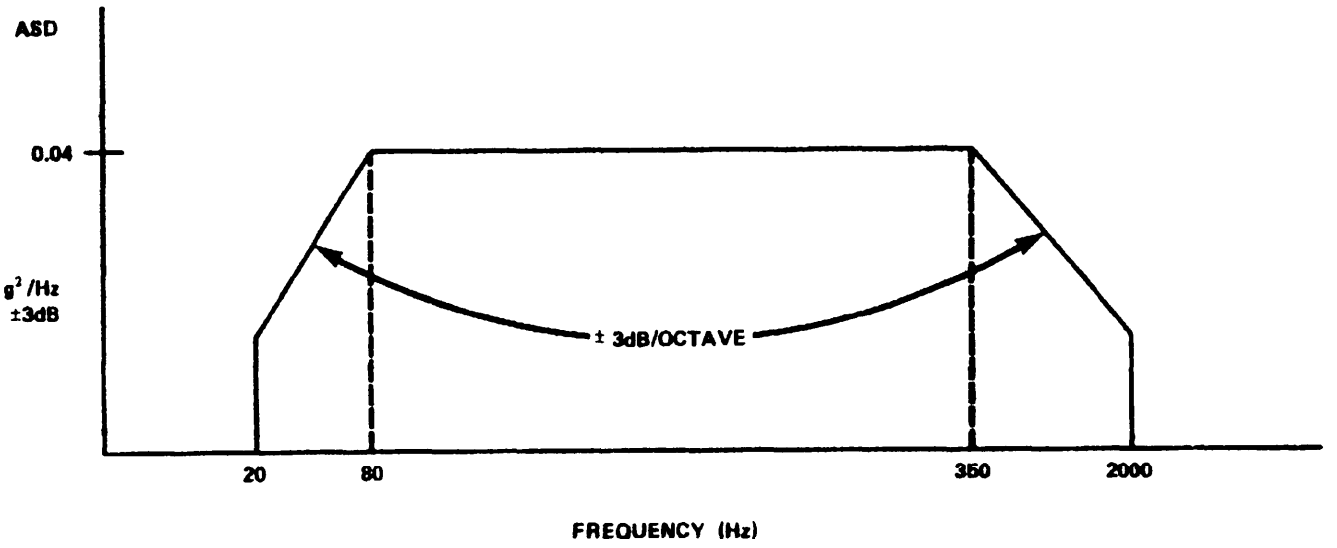


FIGURE 7. Random vibration curve.

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4.5.6.3.3 The test item shall be energized during vibration and appropriate input signals applied to observe any abnormal conditions of the output functional characteristics. All failures occurring during test shall be corrected and the test resumed.

4.5.6.4 Temperature cycling. Each equipment shall be subjected to the temperature curve indicated below. The number of cycles required shall be determined by the electrical and electronic parts count and the parts count chart below. The temperature rate of change shall be not less than 5° Centigrade per minute. Equipment power shall be turned on and off at the indicated times. When practical, the CIU drawers, panels and enclosures shall be opened or removed for maximum exposure to the changing temperature. The last temperature cycle shall be failure free. When performance measurements are called for, a functional operating test shall be performed. The dwell time shall be 80 percent of the time required for the largest electrical or electronic part to become temperature stabilized. Temperature stabilization shall be as defined in MIL-STD-810 for the non operating condition.

Electrical and electronic parts count	No. of temperature cycles
500 or less	3
501 to 1000	4
1001 to 2000	6
2001 to 3000	8
3001 and up	10

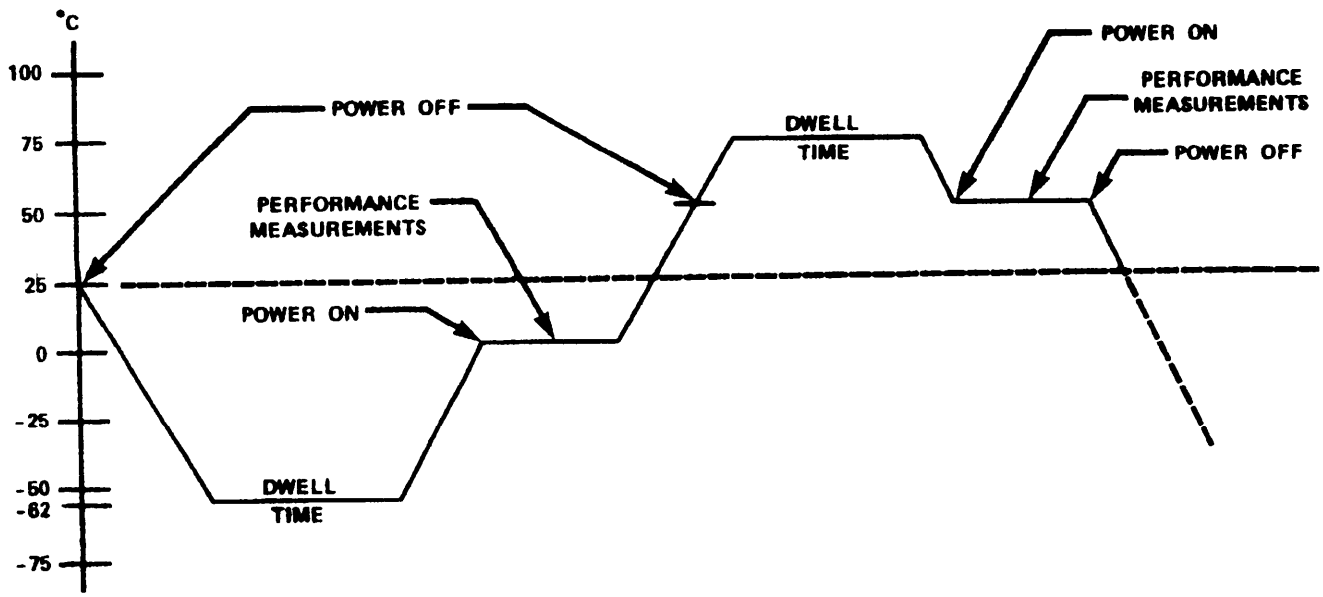


FIGURE 8. Temperature curve (one cycle).

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4.5.7 Electrical tests.

4.5.7.1 Steady-state voltage and frequency test. The CIU shall be operated for at least 15 minutes at each frequency of 50 Hz, 60 Hz, and 400 Hz. The performance readings, including power consumption, shall be measured for each condition as shown in TABLE III.

TABLE III. Steady-state voltage and frequency conditions.

Condition	Voltage			Frequency		
	Lower limit	Normal	Upper limit	Lower limit	Normal	Upper limit
A		X			X	
B	X			X		
C			X	X		
D			X			X

4.5.7.2 Spike voltage test. A spike voltage test shall be conducted to demonstrate the equipment capability to operate normally under severe transient input voltages. A 25-ohm source impedance shall be utilized. The test shall be performed in accordance with the spike voltage test requirement of MIL-E-16400.

4.5.7.3 Hum measurement. The power supply(s) hum shall be measured and recorded at each normal input voltages and frequencies of 50 Hz, 60 Hz, and 400 Hz in accordance with 3.4.1.1.

4.5.7.4 Leakage current test.

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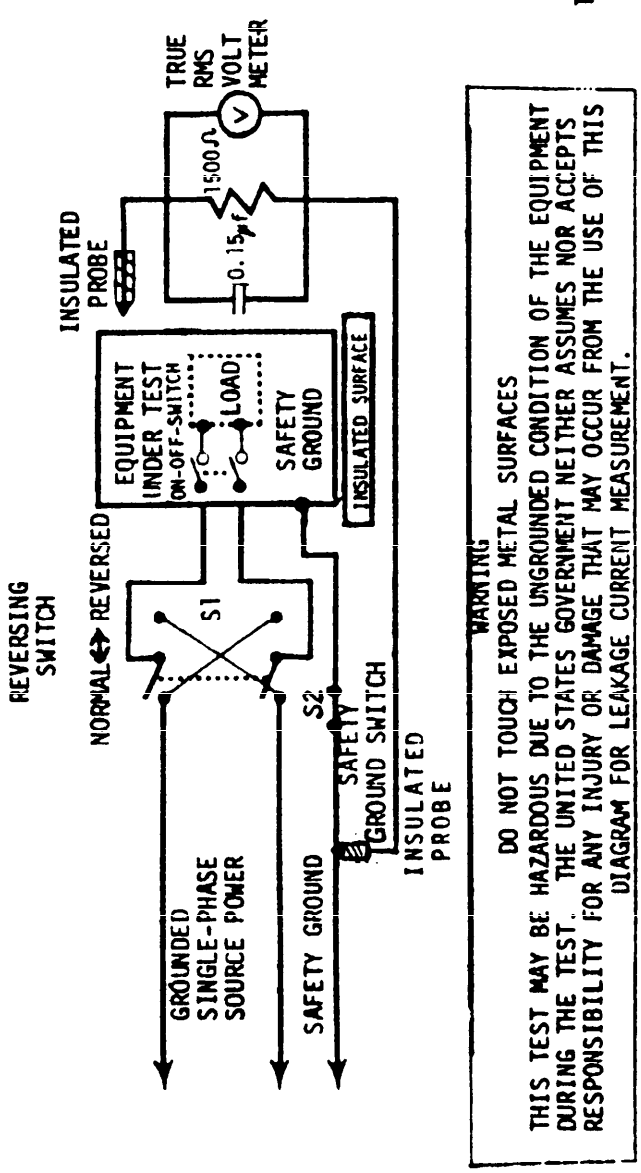
THIS TEST MAY BE HAZARDOUS DUE TO THE UNGROUNDED CONDITION OF THE EQUIPMENT DURING THE TEST. DO NOT TOUCH EXPOSED METAL SURFACES.

THE UNITED STATES GOVERNMENT NEITHER ASSUMES NOR ACCEPTS RESPONSIBILITY FOR ANY INJURY OR DAMAGE THAT MAY OCCUR DURING OR AS A RESULT OF THIS TEST.

4.5.7.4.1 Equipment connections. Each equipment directly connected to an external power source and units deriving power from the equipment shall be placed on an insulated surface. All safety ground conductors between the equipment and units deriving power from the equipment shall be intact. The safety ground conductor between the equipment under test and the source power shall be opened during the test. The equipment shall be connected as shown in FIGURE 9. When DC measurements are performed do not reverse polarity unless the equipment has reverse polarity protection.

4.5.7.4.2 Measurement. Leakage current shall be measured on equipment in its normal operating configuration. Equipment controls in each operating mode shall be such that maximum power will be utilized during leakage current measurements. The leakage current shall be determined by the voltage-drop method. A True RMS Voltmeter shall be used. The voltage measured across the 1500 ohm resistor shall not exceed 7.5 volts at the highest nominal power line voltage and the highest and lowest nominal power line frequencies for which the equipment is designed. The overall measurement error shall not exceed five percent. The probe shall be used on all external conducting parts such as case, connector housings, recessed calibration or adjustment controls, and control shafts with knobs removed, and the voltage measured for every combination of switch positions available in the test diagram. The open safety ground conductor shall be reconnected immediately after the test is completed.

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GENERAL ORDER OF TEST:

1. SOURCE POWER OFF. CONNECT EQUIPMENT PER DIAGRAM.
2. ON-OFF SW OFF. S1 SW NORMAL. S2 SW CLOSED. CONNECT SOURCE POWER.
3. OBSERVE WARNING. S2 SW OPEN. ON-OFF SW ON.
4. FOR EACH PROBE POINT, RECORD VOLTMETER READING (CASE, CONNECTORS, CONTROLS, SHAFTS).
5. ON-OFF SW OFF. REPEAT STEP 4.
6. S1 SW REVERSED.
7. ON-OFF SW ON. REPEAT STEP 4
8. ON-OFF SW OFF. REPEAT STEP 4.
9. S2 SW CLOSED. S1 SW NORMAL.
10. REPEAT STEP 3 THRU 8 FOR EACH MODE OF OPERATION.
11. REMOVE SOURCE POWER. DISCONNECT EQUIPMENT. SIGN RECORD SHEET.

FIGURE 9. Single-phase test diagram for leakage current measurement.

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4.5.8 Electromagnetic characteristics.

4.5.8.1 Electromagnetic environmental effects (E^3) tests. The requirements of 3.4.4 shall be verified by tests performed in accordance with MIL-STD-462. For environmental inspection, only tests CE03 and RE02 shall be performed.

4.5.8.2 Electromagnetic pulse (EMP) hardness validation. The requirements of 3.4.4.1 shall be validated by test, or by analysis which verifies that the CIU is intrinsically EMP-hard or has been hardened.

4.5.9 Human engineering testing. The contractor shall test the CIU for human engineering characteristics in accordance with the Human Engineering in Test and Evaluation paragraph of MIL-H-46855.

4.5.10 Enclosure test. The CIU shall be visually examined for conformance to the dripproof requirement of MIL-STD-108.

4.5.11 Elevation test. The equipment, non-operating, shall be subjected to an equivalent altitude pressure of 15,000 m above sea level for one hour. The elevated pressure shall be returned to normal atmosphere pressure in not more than one minute. A visual inspection shall be made of all component and a one-hour bit error test without noise (see 4.5.4.17) shall be performed after returning the CIU to standard atmospheric pressure.

4.6 Reliability qualification test. Unless otherwise specified (see 6.3) one CIU shall be required for the reliability qualification test. The reliability qualification test shall be performed in accordance with MIL-STD-781, and the accept and reject criteria of Test Plan XVII C. Reliability qualification test conditions shall be in accordance with APPENDIX B, paragraph 50.3 of MIL-STD-781 for sheltered shipboard equipment.

4.7 Maintainability demonstration. The CIU maintainability requirements shall be demonstrated in accordance with Method 9 of MIL-STD-471 by replacement of subassemblies (modules, PC boards) and chassis-mounted electronic, electrical, electromechanical and mechanical components or parts at the organizational level. For the demonstration of mean-corrective maintenance time, a consumer risk, , of 5 percent shall apply. One hundred candidate faults shall be determined in accordance with APPENDIX A of MIL-STD-471 for the demonstration. A representative of the Government shall select a sample of 50 faults from these candidate faults to be used in the demonstration. For fault simulation, a number shall be assigned to each component or part within a subassembly or chassis-mounted component or part. Fault simulation shall be accomplished by introduction of faulty parts, or simulation thereof, corresponding to the number selected.

4.7.1 Demonstration environment. The demonstration at the organizational level shall be performed in a space and temperature environment similar to that in which the equipment will be installed.

4.7.2 Technicians. The demonstration shall be performed by a technician with experience equivalent to that of a third class petty officer. The Government reserves the right to provide and select the technician to perform the equipment maintainability demonstration.

4.7.3 Technical documentation. Technical documentation to be used in the demonstration shall be limited to the technical manual.

4.7.4 Rejection. Failure to meet the acceptance criteria of Method 9 or any of the preceding requirements shall be cause for rejection of the demonstration. In the event of rejection the contractor shall, at no additional cost to the Government, correct all deficiencies and repeat the demonstration until an accept decision is reached.

4.8 Inspection of preparation for delivery. The CIU shall be examined for inclusion of allen wrench (see 3.2.3), proper name, information and warning plates, inclusion of keyline jumper adapter can, mating connectors and backshells, side and top bracket inclusion, intact main and spare fuses and board extenders or cable extenders.

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5. PACKAGING

(The preparation for delivery requirements specified herein apply only for direct Government procurements. Preparation for delivery requirements of referenced documents listed in Section 2 do not apply unless specifically stated in the contract or order. Preparation for delivery requirements for products procured by contractors shall be specified in the individual order.)

5.1 Preservation, packaging, packing, and marking. Unless otherwise specified, preparation for delivery shall be in accordance with the applicable levels of preservation, packaging, packing, and marking specified in MIL-E-17555 (see 6.2).

6. NOTES

6.1 Intended use. The intended use shall be multipurpose for ship and shore as described in the Type A system specification for LINK 11 AN/USQ-74 DTS.

6.2 Ordering data. Procurement documents should specify the following:

6.2.1 Procurement requirements. Procurement documents should specify the following:

a. Title number and date of this specification
b. Quantity of first article sample units required if other than specified in 4.3
c. Government furnished material (see 3.3.9)
d. Interchangeability requirement, if required (see 3.3.3)
e. Levels of preservation, packaging, packing, and marking required (see 5.1)

6.2.2 Data requirements. When this specification is used in a procurement which incorporates a DD Form 1423 and invokes the provisions of 7.104.9(n) of the Armed Services Procurement Regulations (ASPR), the data requirements identified herein will be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved Contract Data Requirements List (DD Form 1423) incorporated into the contract. When the provisions of ASPR 7.104.9(n) are not invoked, the data specified herein will be delivered by the contractor in accordance with the contract requirements. Deliverable data required by this specification are cited in the following paragraphs:

Paragraph	Data requirement	Applicable DID
4.4.1	Test plans	DI-T-3709
4.4.1	Test procedures	UDI-T-22710

(Copies of data item descriptions required by the contractor in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

6.3 First article. When a first article is required, it shall be tested and approved under the appropriate provisions of 7-104.55 of the Armed Services Procurement Regulation. The first article should be a preproduction sample. The first article should consist of at least one unit. The contracting officer should include specific instructions in all procurement instruments, regarding arrangements for examinations, test and approval of the first article.

6.4 Acronyms. Acronyms used in this specification are as follows:

A/D	Analog to digital
ASD	Acceleration Spectral Density
ATE	Automatic test equipment
BER	Bit-error-rate
BIT	Built-in-test
CIC	Combat information center
CIU	Converter Interface Unit
D/A	Digital to analog
DTS	Data terminal set

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EMC	Electromagnetic compatibility
EMI	Electromagnetic interference
GFE	Government furnished equipment
GPETE	General purpose electronic test equipment
HF	High frequency
I/O	Input/output
LSB	Lower side band
MTBF	Mean-time-between-failure
NTDS	Naval Tactical Data System
TMDE	Test measurement and diagnostic equipment
USB	Upper side band

6.5 Definitions. Definitions used in this specification are given in 6.5.1 through 6.5.9.

6.5.1 Aperture time. In a sampling circuit, the time from the sample command to the end of the sampling process when the sample is available.

6.5.2 Failure. The term failure refers to malfunction or parameters deviation that prevents the equipment from performing to full capability in all modes, in the specified operational environment, to operating parameter levels specified herein.

6.5.3 Gain. The ratio of the magnitude (amplitude) of a steady-state sinusoidal output relative to the casual input; the length of a phasor from the origin to a point of the transfer locus in a complex plane.

6.5.4 In-phase signal. An arbitrary designation for one of the outputs of the USB quadrature phase shift circuit, and for one of the outputs of the LSB quadrature phase shift circuit indicating the quadrature phase relation between these two signals and the respective quadrature phase signal outputs of these circuits.

6.5.5 Quadrature. The relation between two periodic functions when the phase difference between them is one-fourth of a period.

6.5.6 Quadrature phase signal. A designation for one of the outputs of USB quadrature phase shift circuit and for one of the outputs of the LSB quadrature phase shift circuit indicating the quadrature phase relation between these two signals and the respective in-phase signal outputs of these circuits.

6.5.7 Saturation. A natural phenomenon or condition in which any further change of input no longer results in appreciable change of outputs.

6.5.8 Settling time. The time required for a D/A converter to be finally within a specified range about its steady-state output for a full scale change in the input signal.

6.5.9 Acceleration spectral density (ASD). Random vibration test.

$$0.04\ g^2/Hz = 6\ g_{rms}$$

6.5.10 Approved Government furnished software. AN/USQ-74 program software, version 7.0 or later, as approved by the Software Life Cycle Support Activity, Software Quality Assurance Management Program.

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